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The Head of the Department
Department of Computer Science and Engineering
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Subject: Submission of the technical proposal "SmartCSE Assist"

Sir/Madam,

I am a student of the Department of Computer Science and Engineering, and I am submitting the enclosed short technical document on **SmartCSE Assist**, a proposed AI-powered mobile application for student academic support. The document is written so that the department can forward it to a software or AI company that may help with development.

The document explains the problem, the proposed system, and its basic workflow (Figure 1). It describes four AI-based features: course recommendation, assignment reminders, basic programming support, and an academic FAQ assistant (Table 1). It also covers the expected benefits, the limitations, and privacy concerns, and it cites published work on recommended systems, AI for education, and data protection.

I believe the application is worth developing for a simple reason. At the moment, students look for academic information in several different places/sites, and advisors answer the same questions again and again. One app that gives personalized course suggestions, deadline reminders, and quick answers could make planning easier for students, and it would collect only a limited amount of student data to do so.

I would be grateful if you could review the proposal and share any suggestions. I am happy to provide more details or to help plan a small pilot version. Thank you for your time and consideration.

Sincerely,

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Student, Department of CSE, BUBT

SmartCSE Assist: An AI-Powered Mobile Application for CSE Student Support

Abstract

Computer Science and Engineering students often collect academic information from notices, reddit/subreddit pages, websites, and messaging groups, which makes planning and deadline tracking difficult. This document proposes SmartCSE Assist, an AI-powered mobile application for academic support. The app offers cacademic planning, course recommendations, assignment reminders, basic programming support, and answers to frequently asked questions related to academics, using limited student data such as courses, academic interests, and activity history. It combines recommended techniques, large language models, and rule-based calculations. The expected benefits are better academic planning, quicker and easier access to information, and fewer frequent queries for advisors.

1. Introduction

1.1 Problem Statement

In many CSE programs, students make academic decisions with scattered information. Course offerings appear on a notice board or student portal, deadlines are shared in class social media groups, and questions about credit rules or prerequisites usually go to academic advisors or senior students. A new student may not know which course matches their interests, and a student with several courses running together can easily miss an assignment date. Programming courses add another problem, since a student stuck on a compile error late at night often has nobody to ask for help. Reviews of AI in higher education list profiling, adaptive systems, and intelligent tutoring among the main application areas [1], [2], so these problems are suited to an AI-based solution.

1.2 AI-Based Features

SmartCSE Assist includes four AI-based features. First, a **course recommendation module** suggests courses and electives using a student's completed courses, stated interests, and activity history. Recommendation systems are widely used for this kind of personalized filtering [3], and matrix factorization is a common technique when enough interaction data exists [4]. Second, an **assignment reminder module** collects deadlines and orders them by urgency and workload. Third, a **programming support assistant** explains compiler errors and gives hints for basic C, C++, and Python problems; language models trained on code can write short programs from natural human-language descriptions [6], so explaining and hinting on small problems is realistic. Fourth, an **academic FAQ assistant** answers routine questions about rules, schedules, and procedures from approved department documents. It uses retrieval-augmented generation so that each answer stays tied to those documents [5]. Table 1 in Section 4 summarizes these features.

2. Proposed System

SmartCSE Assist will be a mobile application for Android and iOS, with a cloud backend that hosts the AI services. A student signs in with a university/student account and gives consent

for the data the app may use. The app stores only limited information: enrolled and completed courses, interests selected by the student, and basic activity history such as the features used and the recommendations accepted. Personal identifiers are kept separate from the data used by the models. Since student records are personal data, the design follows the data minimization and consent principles of the GDPR [7] as a reference framework.

3. System Workflow

The basic workflow is shown in Figure 1. The student sends a request through the mobile app, for example by asking for elective suggestions or typing a programming question. The request goes to the back-end API, which checks login and consent and passes only the required data to the AI engine. The AI engine has three parts: a recommender for courses, a language model for programming help and FAQs, and a scheduling component for reminders. These parts read from the data layer, which holds the student profile, the course catalog, and the approved FAQ documents. The result returns to the app as a suggestion, a reminder, or a text answer. The student can rate a response, and this feedback is stored so that the recommender can be updated later.

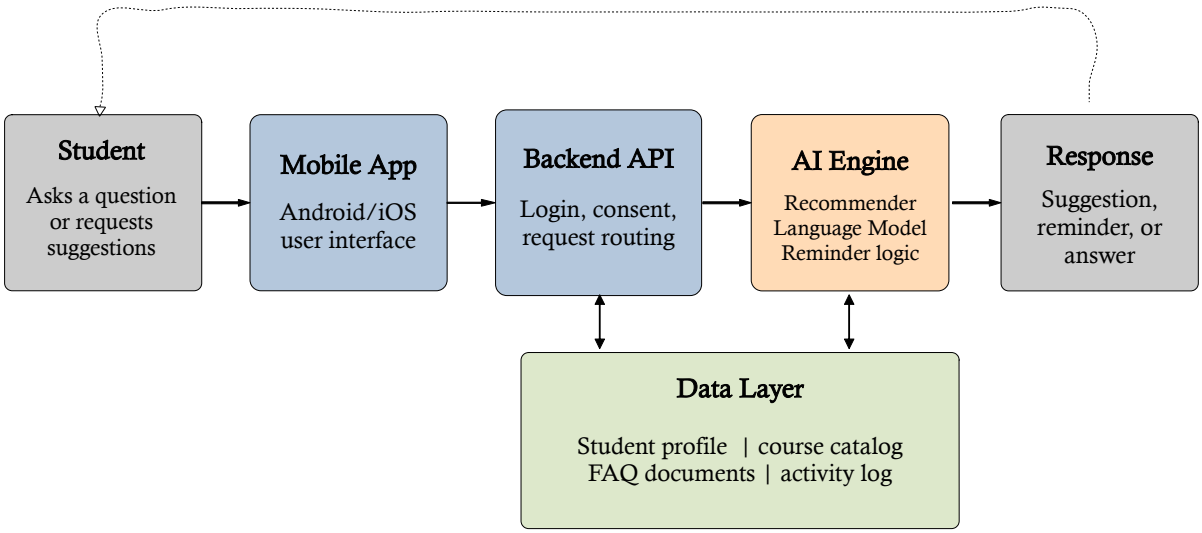

Figure 1. Proposed workflow and architecture of SmartCSE Assist.

4. Feature Details

Table 1 compares the four AI-based features by input data, AI technique, and expected output. The techniques are common choices and may change after prototype testing. For example, the course recommendation will start with content-based filtering, because a new student has little activity history (the cold-start problem), and collaborative filtering can be added as more data is collected [3].

***Table 1**: AI-based features of SmartCSE Assist and their expected benefits.*

Feature	Input / data	AI technique	Expected output / benefit
Course recommendation	Completed courses, interests, activity history, course catalog	Content-based and collaborative filtering; matrix factorization	Ranked list of suggested courses with a short reason; easier elective choice

Feature	Input / data	AI technique	Expected output / benefit
Assignment reminders	Deadlines, student's course list, past submission activity	Rule-based scheduling with a priority score	Timely notifications ordered by urgency; fewer missed deadlines
Programming support	Code snippet, error message, selected topic	Language model trained on code, prompted to give hints	Error explanation and step-by-step hints for basic problems
Academic FAQ assistant	Student question, approved department documents	Natural language processing with retrieval-augmented generation	Quick answer with its source document; fewer routine queries to advisors

5. Expected Benefits and Limitations

5.1 Expected Benefits

Students will have one place for planning, reminders, and quick answers. Advisors and instructors may spend less time on repeated questions. Personalized suggestions can also help students notice elective courses they would not have found on their own. These are expected benefits only. Their size would have to be measured in a pilot using student feedback and usage data.

5.2 Limitations

The system has some clear limitations. Language models can give wrong or misleading answers, so the FAQ assistant should answer only from approved documents and show its source, and the programming assistant should give hints instead of full solutions. Recommendation quality depends on the data, and new students will have little history. Deadlines may need to be entered by students, because the app will not have access to every course page. Even limited academic data is sensitive, so privacy and security need regular attention. Finally, the app does not replace faculty advising. Final decisions on course registration stay with the student and the advisor.

6. Conclusion

SmartCSE Assist is proposed to reduce the effort students spend on planning, tracking deadlines, and finding routine academic information. It combines four AI-based features with limited data collection. The next steps are a small prototype for one batch, an evaluation of recommendation quality and answer accuracy with student feedback, and a privacy review before wider use. Help from a software or AI company would be useful for model integration, mobile development, and secure hosting.

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